

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Chapter 3

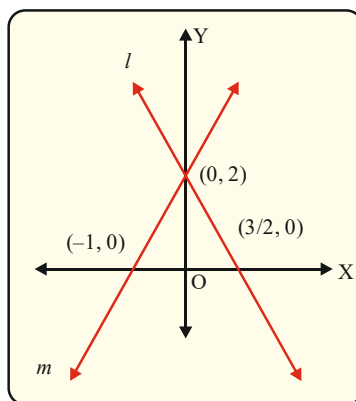


Exercise-1

Part - 1 (Basic Multiple Choice Questions)

1. The system of linear equations represented by the lines l and m is

[CBSE SQP (Basic) 2024-25]



- (a) Consistent with unique solution
(b) Inconsistent
(c) Consistent with three solutions
(d) Consistent with many solutions
2. If a pair of linear equations is consistent, then the lines are
(a) Parallel
(b) Always coincident
(c) Intersecting or coincident
(d) Always intersecting
3. The pair of equations $y = 0$ and $y = -7$ has
(a) One solution
(b) Two solutions
(c) Infinitely many solutions
(d) No solution.
4. One equation of a pair of dependent linear equations is $-5x + 7y = 2$. The second equation can be
(a) $10x + 14y + 4 = 0$
(b) $-10x - 14y + 4 = 0$
(c) $-10x + 14y + 4 = 0$
(d) $10x - 14y = -4$
5. Two linear equations in variables x and y are below:
 $a_1x + b_1y + c = 0$
 $a_2x + b_2y + c = 0$

Which of the following pieces of information is independently sufficient to determine if a solution exists or not for this pair of linear equations?

[CBSE APQ (Standard) 2023-24]

- (i) $\frac{a_1}{b_1} = \frac{a_2}{b_2} = 1$ (ii) $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ (iii) $\frac{a_1}{a_2} = \frac{a_1}{b_1} \neq 1$ (iv) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$
 (a) iv only (b) i and iv (c) ii and iv (d) i and iii

6. If $3x - 4y = 6$ and $y = \frac{2x}{3}$, then the value of x is
 (a) 12 (b) 14 (c) 16 (d) 18
7. The sum of the digits of a two-digit number is 9. If 27 is added to it, the digits of the number get reversed. The number is
 (a) 25 (b) 72 (c) 63 (d) 36
8. Rahul's house has rectangular garden in front of it for children to play in. The length of the rectangle exceeds its width by 6 m and its perimeter is 60m. If length is x and width is y, then the dimensions of the garden are :
 (a) $x = 12\text{m}, y = 18\text{m}$ (b) $x = 33\text{m}, y = 27\text{m}$ (c) $x = 18\text{m}, y = 12\text{m}$ (d) $x = 15, y = 22\text{m}$
9. The value of k for which the pair of linear equations $3x + 2y = -5$ and $x - ky = 2$ has unique solution is :

[Delhi Gov. SQP 2022]

- (a) $R - \left\{ \frac{-2}{3} \right\}$ (b) $R - \left\{ \frac{-1}{3} \right\}$ (c) $R - \left\{ \frac{-2}{5} \right\}$ (d) $R - \left\{ \frac{1}{2} \right\}$

10. Aruna has only Rs 1 and Rs 2 coins with her. If the total number of coins that she has is 50 and the amount of money with her is Rs 75, then the number of Rs 1 and Rs 2 coins respectively, are : **[Mod. NCERT Exemplar]**



- (a) 35 and 15 (b) 35 and 20 (c) 15 and 35 (d) 25 and 25
11. The pair of equations $ax + 2y = 9$ and $3x + by = 18$ represent parallel lines, where a, b are integers, if :
 (a) $a = b$ (b) $3a = 2b$ (c) $2a = 3b$ (d) $ab = 6$

Part - 2 (Basic Subjective Type Questions)

- Do the following pair of linear equations have no solution?
Justify your answer.
(i) $5x + 6y = 30$ $10x + 12y = 40$
(ii) $x = 2y$ $y = 2x$
(iii) $27x + 3y = 7$ $3x + \frac{9}{2}y = 12$
- Find c if the system of equations $cx + 3y + (3 - c) = 0$; $12x + cy - c = 0$ has infinitely many solutions
[CBSE 2019]
- For which value(s) of λ do the pair of linear equations $\lambda x + y - \lambda^2 = 0$ and $x + \lambda y - 1 = 0$ have :
[NCERT Exemplar]
(a) no solution? (b) infinitely many solutions? (c) a unique solution?
- Solve for x and y : $\frac{x+1}{2} + \frac{y-1}{3} = 8$ and $\frac{x-1}{3} + \frac{y+1}{2} = 9$
- A two digit number is 4 times the sum of its digits. It is also equal to 3 times the product of its digits. Find the number.
[CBSE 2016]
- For what value of p will be the following system of equations represent coincident lines?
 $x + 5y = 7$
 $3x + 15y = p$
- A lady has 25p and 50p coins in her purse. If in all she has 40 coins totalling Rs. 12.50, find the number of coins of each type she has.

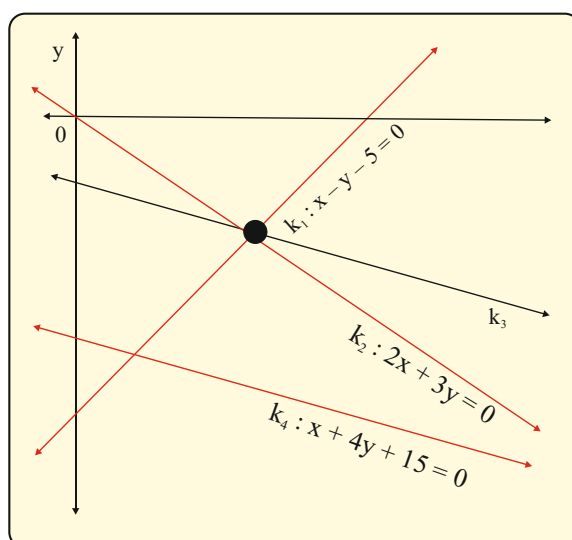


Exercise-2

Part - 1 (Advanced Multiple Choice Questions)

- Aditya is walking along the line joining (1, 4) and (0, 6). Aditi is walking along the line joining (3, 4) and (1, 0). Find the point where both of them cross each other.
(a) (1, 2) (b) (2, 1) (c) (2, 2) (d) (2, 3)
- If we draw the graphs of the following pair of linear equations : $x + 3y = 6$ and $2x - 3y = 12$, then the ratio of the areas of the two triangles formed by first line, $x = 0$, $y = 0$ and second line, $x = 0$, $y = 0$ is
(a) 1 : 2 (b) 2 : 1 (c) 2 : 3 (d) 3 : 2
- If we draw the graphs of the equations $x = 3$, $x = 5$ and $2x - y - 4 = 0$, then the area of the quadrilateral formed by the lines and the X-axis is
(a) 4 sq. units (b) 8 sq. units (c) 16 sq. units (d) 20 sq. units
- Harsh correctly solved a pair of linear equations in two variables and found their only point of intersection as (3, -2). One of the lines was $x - y = 5$. Which of the following could have been the other line?
[CBSE Competency Focused Practice Questions 2022-23]
(i) $3x - 3y = 15$ (ii) $2x - 3y = 12$ (iii) $2x - 3y = 14$
(a) only I (b) only II (c) only I and II (d) only II and III

5. A team of students is formed by including the students from various divisions of class X. The number of students in the team is 8 times the number of divisions and the sum of the number of divisions and the number of students in the team is 27. Find the number of students in the team.
(a) 24 (b) 32 (c) 12 (d) 25
6. If $ax + by = a^2 - b^2$ and $bx + ay = 0$, then the value of $x + y$ is [CBSE (Standard) 2024]
(a) $a^2 - b^2$ (b) $a + b$ (c) $a - b$ (d) $a^2 + b^2$
7. A shopkeeper sells a saree at 8% profit and a sweater at 10% discount, thereby, getting a sum Rs. 1008. If she had sold the saree at 10% profit and the sweater at 8% discount, she would have got Rs. 1028. Find the cost price of the saree.
(a) Rs 400 (b) Rs 500 (c) Rs 600 (d) Rs 550
8. Which of the following statements is NOT true?
(a) If a pair of equations has exactly one solution, it is consistent.
(b) If a pair of equations has infinitely many solutions, it is consistent.
(c) If a pair of equations has no solution, it is inconsistent.
(d) If a pair of equations has infinitely many solutions, it is inconsistent
9. The values of x and y satisfying the two equations $32x + 33y = 34$, $33x + 32y = 31$ respectively are : [CBSE Term - 1 Std. 2021]
(a) $-1, 2$ (b) $-1, 4$ (c) $1, -2$ (d) $-1, -4$
10. In a $\triangle ABC$, $\angle A = x^\circ$, $\angle B = (3x - 2)^\circ$, $\angle C = y^\circ$. Also $\angle C - \angle B = 9^\circ$. The sum of the greatest and the smallest angle of this triangle is : [CBSE Term - 1 Std. 2021]
(a) 107° (b) 135° (c) 155° (d) 145°
11. The ratio of a two - digit number and the sum of its digits is 7:1. How many such two - digit numbers are possible?
(a) 1 (b) 4 (c) 9 (d) (Infinitely many)
12. Shown below is a graph with four straight lines. It is given that k_1 , k_2 and k_3 intersect at exactly one point and line $k_3 \parallel k_4$.



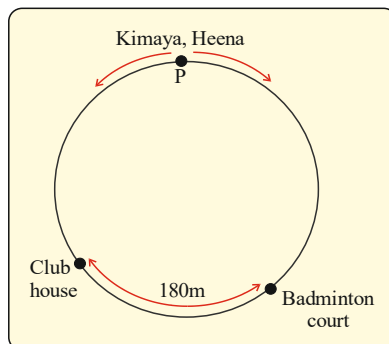
Which of the following is the equation of line k_3 ?

- (a) $x + y - 1 = 0$ (b) $x + 4y + 5 = 0$ (c) $x - 4y - 11 = 0$ (d) $2x + 8y + 35 = 0$

13. Shown below is a pair of linear equations.
 $mx + 4y - 6 = 0$
 $ny - 12x + 12 = 0$
 The values of m and n are _____ and _____ respectively for the equations to have infinitely many solutions.
 (a) $-1, 2$ (b) $-1, 3$ (c) $6, -8$ (d) $6, -2$
14. A can do a piece of work in 24 days. If B is 60% more efficient than A , then the number of days required by B to do the twice as large as the earlier work is
 (a) 24 (b) 36 (c) 15 (d) 30
15. X 's salary is half that of Y 's. If X got a 50% rise in his salary and Y got 25% rise in his salary, then the percentage increase in combined salaries of both is
 (a) 30 (b) $33\frac{1}{3}$ (c) $37\frac{1}{2}$ (d) 75
16. At present ages of a father and his son are in the ratio 7 : 3, and they will be in the ratio 2 : 1 after 10 years. Then the present age of father (in years) is
 (a) 42 (b) 56 (c) 70 (d) 77
17. A motor boat takes 2 hours to travel a distance 9km down the current and it takes 6 hours to travel the same distance against the current. The speed of the boat in still water and that of the current (in km/hour) respectively are
 (a) 3, 1.5 (b) 3, 2 (c) 3.5, 2.5 (d) 3, 1
18. The 2 digit number which becomes $(5/6)$ th of itself when its digits are reversed. The difference in the digits of the number being 1, then the two digits number is
 (a) 45 (b) 54 (c) 36 (d) None of these
19. x and y are 2 different digits. If the sum of the two digit numbers formed by using both the digits is a perfect square, then value of $x + y$ is
 (a) 10 (b) 11 (c) 12 (d) 13
20. If $3x + 4y : x + 2y = 9 : 4$, then $3x + 5y : 3x - y$ is equal to
 (a) 4 : 1 (b) 1 : 4 (c) 7 : 1 (d) 1 : 7
21. A boat travels with a speed of 15km/hr in still water. In a river flowing at 5km/hr, the boat travels some distance downstream and then returns. The ratio of average speed to the speed in still water is
 (a) 8 : 3 (b) 3 : 8 (c) 8 : 9 (d) 9 : 8
22. For what value of p , the following pair of linear equations in two variables will have infinitely many solutions?
 $px + 3y - (p - 3) = 0$, $12x + py - p = 0$
 (a) 6 (b) -6 (c) 0 (d) 2
23. In a classroom, one-fifth of the boys leave the class and the ratio of the remaining boys to girls is 2 : 3. If further 44 girls leave the class, then the ratio of boys to girls is 5 : 2. How many more boys should leave the class so that the number of boys equals that of girls?
 (a) 16 (b) 24 (c) 30 (d) 36

Part - 2 (Advanced Subjective Type Questions)

1. Kimaya and Heena started walking from the point P at the same moment in opposite directions on a 800 m long circular path as shown in figure. Kimaya walked to the club house at an average speed of 100m/min and Heena walked to the badminton court at an average speed of 80 m/min. The length of the circular track between the club house and the badminton court is 180 m. **[CBSE Practice Set-1, 2023]**



(Note : The figure is not to scale)

If Heena took 1 minute more than Kimaya to reach her destination, find the time taken by Heena to reach the badminton court. Show your work.

2. The area of a rectangle reduces by 160 m^2 if its length is increased by 5m and breadth is reduced by 4m. However, if the length is decreased by 10m and breadth is increased by 2m, then its area is decreased by 100 m^2 . Find the dimensions of the rectangle. **[CBSE Practice Set-2 2023]**
3. Given below is a pair of linear equations : **[CBSE Practice Set-1 2023]**

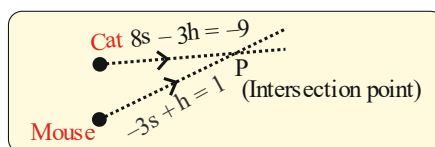
$$2x - my = 9$$

$$4x - ny = 9$$

Find at least one pair of the possible values of m and n, if exists, for which the given pair of linear equations has :

- (a) A unique solution
(b) Infinitely many solutions
(c) No solution.
4. A man wished to give Rs 12 to each person and found that he fell short of Rs 6 when he wanted to give to all the persons present. He therefore, distributed Rs 9 to each person and found that Rs 9 was left over. How much money did he have and how many persons were there? **[Disha Govt.]**
5. A computer animation below shows a cat moving in a straight line

Its height, h metres, above the ground, is given by $8s - 3h = -9$, where s is the time in seconds after it starts moving. In the same animation, a mouse starts to move at the same time as the cat and its movement is given by $-3s + h = 1$.



- (a) Draw the graph of the two equations on the same sheet of graph paper.
(b) Will the cat be able to catch the mouse?
(c) If yes, after how much time and at what height will the cat catch the mouse?

6. If 16 is subtracted from twice the greater of two positive numbers, the result is half the other number. If 1 is subtracted from half the greater number, the result is still half the other number. Find the two numbers.
[CBSE Practice Set-2 2023]
7. Places A and B are 180 km apart on a highway. One car starts from A and another from B at the same time. If the car travels in the same direction at different speeds, they meet in 9 hours. If they travel towards each other with the same speeds as before, they meet in an hour. What are the speeds of the two cars?
[CBSE SQP 2024]
8. Vijay had some bananas and he divided them into two lots A and B. He sold the first lot at the rate of Rs. 2 for 3 bananas and the second lot at the rate of Rs 1 per banana and got a total of Rs 400. If he had sold the first lot at the rate of Rs 1 per banana and the second lot at the rate of Rs 4 for 5 bananas, his total collection would have been Rs 460. Find the total number of bananas he had.
[Delhi Gov QB 2022]
9. Solve the following system of linear equations :
- $$2(ax - by) + (a + 4b) = 0$$
- $$2(bx + ay) + (b - 4a) = 0$$
10. A vessel contains mixture of 24 Ltr milk and 6 Ltr water and a second vessel contains a mixture of 15 Ltr. milk and 10Ltr. water.
How much mixture of milk and water should be taken from the first and the second vessel separately and kept in a third vessel so that the third vessel may contain a mixture of 25 Ltr. milk and 10 Ltr. water?
11. The sum of a two-digit number and the number obtained by reversing the digits is 66. If the digits of the number differ by 2, find the number. How many such numbers are there?
12. A man travels 600 km partly by train and partly by car. If he covers 400 km by train and the rest by car, it takes him 6 hours and 30 minutes. But if he travels 200 km by train and rest by car, he takes half an hour longer. Find the speed of the train and that of the car.
13. A two digit number is obtained by either multiplying sum of the digits by 8 and adding 1 or by multiplying the difference of the digits by 13 and adding 2. Find the number.
14. The sum of a two digit number and the number obtained by reversing the order of its digits is 99. If the digits differ by 3, find the number.
15. The numerator of a fraction is 4 less than the denominator. If the numerator is decreased by 2 and the denominator is increased by 1, then the denominator is eight time the numerator. The fraction is $\frac{a+b}{2}$. Find $\frac{a+b}{2}$.

Part - 3 (Advanced Miscellaneous Questions)

- Directions :** In the following questions, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:
 - (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
 - (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
 - (c) Assertion (A) is true but reason (R) is false.
 - (d) Assertion (A) is false but reason (R) is true.
1. **Assertion (A) :** If lines $mx + 4y - 6 = 0$ and $ny - 12x + 12 = 0$ have infinitely many solutions, then $m = 1$ and $n = -3$.

Reason (R) : If lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ has infinitely many solutions, then $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

2. **Assertion (A) :** For $\lambda = 1$, the pair of linear equations $\lambda x + y = \lambda^2$ and $x + \lambda y = 1$ has no solution.

Reason (R) : The pair of linear equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ has no solution, if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

3. **Assertion (A) :** $9x + 12y - 7 = 0$ and $6x + 8y - 14 = 0$ form a consistent pair of linear equations.

Reason (R) : A pair of linear equations, $px + qy + r = 0$ and $fx + gy + h = 0$, has no solution if $\frac{p}{f} = \frac{q}{g} \neq \frac{r}{h}$.

4. Equations are given in column I. Match the following with solutions and conditions mentioned in column II and choose the correct option.

Column - I	Column - II
(A) $3x + 4y = 15, 4x + y = 16$	(I) $x = 2, y = -3$; consistent
(B) $6x + y + 3 = 0, 4x + 4y = 8$	(II) $x = 1, y = 3$; consistent
(C) $x + y - 3 = 0, 4x + 4y = 12$	(III) $x = -1, y = 3$; consistent
(D) $2x + y - 1 = 0, x + 4y + 10 = 0$	(IV) $x = -2, y = 5$; dependent, consistent

- (a) A-IV, B-I, C-II, D-III
(b) A-II, B-III, C-IV, D-I
(c) A-IV, B-III, C-II, D-I
(d) A-III, B-I, C-IV, D-II
5. Solutions are given in Column I. Match the following with a pair of equations in Column II and choose the correct option.

Column - I	Column - II
(A) $x = 3, y = -2$	(I) $x + y = 8, x - y = 2$
(B) $x = 5, y = 3$	(II) $x = y, x - y = 4$
(C) $x = 2, y = -2$	(III) $x = y, x - y = 2$
(D) $x = 1, y = -1$	(IV) $x + y = 1, x - y = 5$

- (a) A-IV, B-I, C-II, D-III
(b) A-III, B-I, C-II, D-IV
(c) A-IV, B-III, C-I, D-II
(d) A-III, B-IV, C-I, D-II
6. The system of linear equations is given in Column I. Match the following with certain conditions mentioned in Column II and choose the correct option.

Column - I	Column - II
(A) i. $3x - 4y = 8$ ii. $2x - 2y = 6$ iii. $4x - 5y = 11$	(I) I and II are parallel lines, I and III are intersecting lines
(B) i. $6x - 3y = 3$ ii. $3x + y = 4$ iii. $6x + 2y = 5$	(II) I, II and III intersect at a common point
(C) i. $4x - 7y = 18$ ii. $8x - 14y = 36$ iii. $12x - 21y = 45$	(III) I and II are intersecting lines, II and III are parallel lines.
(D) i. $8x - y = 14$ ii. $40x - 5y = 50$ iii. $5x - 4y = 20$	(IV) I and II are coincident lines, I and III are parallel lines

(a) A-IV, B-III, C-I, D-II

(b) A-III, B-I, C-IV, D-II

(c) A-IV, B-III, C-II, D-I

(d) A-II, B-III, C-IV, D-I

7. If a pair of linear equations is given by $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ and $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$. In this case, the pair of linear equations is consistent. (True / False)
8. If a pair of linear equations is given by $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ and $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$. In this case, the pair of linear equations is consistent. (True / False)
9. If a pair of linear equations is given by $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ and $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$. In this case, the pair of linear equations is consistent. (True / False)
10. If a $\triangle ABC$, $\angle C = 3 \angle B = 2(\angle A + \angle B)$, then angles are $20^\circ, 40^\circ, 100^\circ$. (True / False)



Exercise-3

(Case Based Questions)

1. Sarita was attempting a test consists of 'Multiple Choice Questions'. One mark is awarded for every correct answer while $\frac{1}{4}$ mark is deducted for every wrong answer. Sarita knew answers to some of the questions. Rest of the questions she attempted by guessing. She answered 120 questions.



- (i) Sarita knew answers to 80 questions and if half of the answers she attempted by guessing were correct, then how many marks she got?
- (ii) If answer to all questions she attempted by guessing were wrong and answered 80 correctly, then how many marks she got?
- (iii) If answers to all questions she attempted by guessing were wrong, then how many questions did she answer correctly to score 90 marks?

OR

- (iii) If answer to all questions she attempted by guessing were wrong, then how many questions answered correctly to score 95 marks?

2. The total cost of snowden ice cream parlour is divided into fixed cost (x) and variable cost (y). Fixed cost is the cost that the ice cream parlour has to incur even at zero level of production and variable cost is the cost that will be directly proportional to each unit of ice cream sold.

The parlour launched a new flavour of ice cream and wanted to find the fixed and variable cost associated with it. They found that their total cost for the flavour was Rs 27,500/- after selling 150 units and Rs. 32,500/- after selling 250 units.



- (i) Frame the equations that represent the total cost incurred by Snowden ice cream parlour for the flavour in terms of fixed and variable costs
- (ii) Shyam calculated the fixed cost as Rs 2000. Check if his answer is right or not.
- (iii) Calculate the fixed cost incurred by the ice-cream parlour for the new flavour

OR

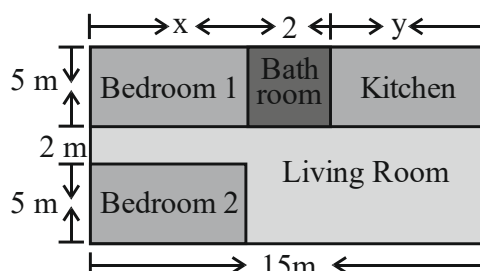
- (iii) Calculate the total variable cost for 20 ice-creams incurred by the ice-cream parlour for the new flavour.

3. A test consists of 'True' or 'False' questions. One mark is awarded for every correct answer while $\frac{1}{4}$ mark is deducted for every wrong answer. A student knew answers to some of the questions. Rest of the questions he attempted by guessing. He answered 120 questions. **[Mod. CBSE Question Bank 2022]**

Types of Question	True/False
Marks given for correct answer	1
Marks deducted for wrong answer	0.25

- (a) If answer to all questions he attempted by guessing were wrong then how many questions did he answer correctly to score 90 marks?
 - (a) 90
 - (b) 96
 - (c) 100
 - (d) 80
- (b) If answer to all questions he attempted by guessing were wrong and he answered 80 correctly, then how many marks did he get?
- (c) If answer to all questions he attempted by guessing were wrong; then how many questions did he answer correctly to score 95 marks?
 - (a) 90
 - (b) 75
 - (c) 100
 - (d) 80

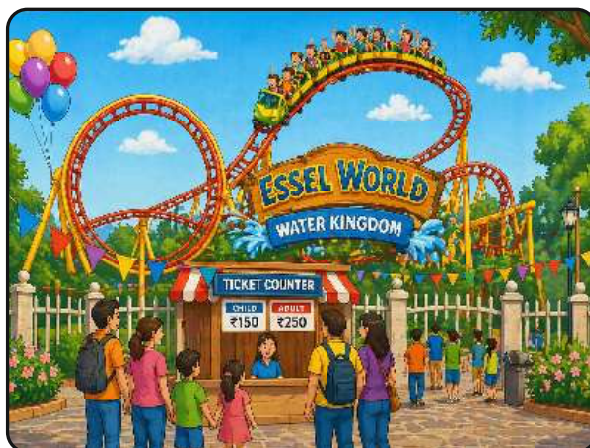
4. Amit is planning to buy a house and the layout is given below. The design and the measurement have been made such that the area of the two bedrooms and kitchen together is 95 sq. m.



- (a) Write the pair of linear equations to depict the above situation.
- (b) (i) What is the length of the outer boundary of the layout

OR

- (ii) What is the area and cost of laying tiles in the living room at the rate of Rs 50 per sq. m of the layout?
 - (c) What is the cost of laying tiles in the kitchen at the rate of Rs 50 per sq.m of the layout?
5. Essel World is one of India's largest amusement parks that offers a diverse range of thrilling rides, water attractions and entertainment options for visitors of all ages. The park is known for its iconic "Water Kingdom" section, making it a popular destination for family outings and fun-filled adventure. The ticket charges for the park are Rs. 150 per child and Rs. 250 per adult.



On a day, the cashier of the park found that 300 tickets were sold and an amount of Rs, 55,000/- was collected.

Based on the above, answer the following questions :

- (a) If the number of children visited be x and the number of adults visited by y , then write the given situation algebraically.
- (b) (i) How many children visited the amusement park that day?
(ii) How many adults visited the amusement park that day?
- (c) How much amount will be collected if 250 children and 100 adults visit the amusement park?

Answer Key



Exercise-1

Part - 1 (Basic Multiple Choice Questions)

- (1) A (2) C (3) D (4) D (5) B (6) D
(7) D (8) A (9) A (10) D (11) D

Part - 2 (Basic Subjective Type Questions)

- (1) (i) No solution (ii) doesn't satisfy the condition (iii) doesn't satisfy
(2) The value of c is 6
(3) (a) $\lambda = -1$ (b) $\lambda = 1, 0$ (c) $\lambda \neq \pm 1$
(4) $x = 7, y = 13$
(5) The number is 24
(6) $P = 21$
(7) $x = 30$ (25 p coins) & $y = 10$ (50 p coins)



Exercise-2

Part - 1 (Advanced Basic Multiple Choice Questions)

- (1) C (2) A (3) B (4) B (5) A (6) C
(7) C (8) D (9) A (10) A (11) B (12) B
(13) C (14) D (15) B (16) C (17) A (18) B
(19) B (20) C (21) C (22) A (23) B

Part - 2 (Advanced Subjective Type Questions)

- (1) The times taken by Heena is 4 minutes
(2) $l = 60\text{m}$ & $b = 20\text{m}$
(3) (a) $m = 1, n = 1$ (b) No such value possible (c) $m = 1, n = 2$
(4) He has Rs 54 and there were 5 persons
(5) (a) Graph of 2 lines intersecting at (6, 19) (b) yes (c) $s = 6$ sec, $h = 14$ meters
(6) The two numbers are 10 and 8

- (7) $x = 100 \text{ km/hr}$, $y = 80 \text{ km/hr}$
- (8) ($x = 300$, $y = 200$) The total number of bananas he had is 500
- (9) $x = -1/2$, $y = 2$
- (10) $x = 20$, $y = 15$
- (11) The number is 24. Thus, there are two such numbers 42 and 24.
- (12) Speed of train = 100 km/h , Speed of Car = 80 km/h
- (13) The number is 41
- (14) The number is 36 or 63
- (15) 5

Part - 3 (Advanced Miscellaneous Questions)

- (1) D (2) D (3) D (4) B (5) A (6) D
- (7) True (8) False (9) True (10) False



Exercise-3

(Case Based Questions)

- (1) (i) 95 Marks (ii) 70 Marks (iii) $x = 96$, $y = 24$ **OR** (iii) $x = 100$, $y = 20$
- (2) (i) Required equations are $x + 150y = 27,500$, $x + 250y = 32,500$
- (ii) Shyam is wrong (iii) Rs 20,000/- **OR** (iii) Rs. 1,000/-
- (3) (a) B (b) 70 marks (c) C
- (4) (a) $x + y = 13$ (b) (i) 54 m **OR** (ii) Rs. 3750/- (c) Rs. 1750/-
- (5) (a) $x + y = 300$ (b) (i) 200 (ii) 100 (c) Rs. 62,500/-